MIL-O-60831B(MU)
9 August 1973
SUPERSEDING
MIL-O-60831A(MU)
10 April 1968

MILITARY SPECIFICATION

OBTURATING BAND, PLASTIC, SPLIT (MOLDED)

1. SCOPE

1.1 This specification covers Obturating Band, Plastic, Split (Molded) for 81MM, M374, H.E., and 81MM, M375, WP, Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

Federal Test Method Standard No. 406 - Plastic, Methods of Testing

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

C10534925	- Ring, Obturating for 81MM, Mortar
IEL10534925	- Inspection Equipment List for Ring, Obturating for 81MM, Mortar

FSC 1395

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American National Standards Association, 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 General.-All parts shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 Material.-The material shall be as specified on the applicable drawing. No regrind or scrap will be allowed.

3.2.1 Certification.-Certification must be furnished by the molder for each lot of bands certifying that the plastic used in molding complies with the requirements of 3.2.

3.3 Molded Parts.-Parts made in accordance with the applicable drawings shall be produced by the injection molding process. Unless otherwise specified, all tests required by this document shall be performed under the standard conditions and on specimens which have been conditioned in accordance with the requirements of Federal Test Methods Standard No. 406.

3.3.1 Workmanship.-The molded parts shall be uniform in color, free of cracks, burn marks, flashing, contamination, sprues and gates.

3.4 Hidden Defects.-When parts which have been inspected, tested and accepted, contain inherent defects subsequently exposed, the contractor may be required to replace the defective material without expense to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 100 bands. The sample shall equally represent all mold cavities.

4.2.1.1 Examination and Test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.3 Inspection.

4.3.1 Lot

4.3.1.1 Submission of Product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Band Lot.-A lot shall consist of bands produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Each lot shall contain: bands manufactured from one batch or "run" of resin and molded in one mold on one machine.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect shall be used. Sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during first article testing and after each 1000 items produced from each cavity mold for each machine in operation and after any tooling change. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

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4.3.2.2 Ring Drawing C10534925.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Thickness	Gage
102. Width	Gage
103. I.D. (Functional)	Gage
104. Radius of corners on outside diameter, max.	Gage
MINOR	
201. Overlap	Visual

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packing, Level C.-Preservation and packaging of the bands shall be in accordance with the requirements of MIL-STD-1169. In addition:

1. Bands shall be placed on cardboard tubes, the dimensions of which are as follows:

Length - 20 inches nominal
Thickness - 0.10 inches nominal
Outer diameter - 2 7/8 inches ± 0.015

The tubes shall be smooth with no delaminations, breaks or bends on the outside surface.

2. The bands shall be placed on the tubes such that no overlapping of bands shall occur.

3. Ninety to one-hundred bands shall be placed on each tube and the tubes with bands shall be placed into individual polyethylene bags and sealed.

4. The tubes shall then be placed into the compartments of a corrugated cardboard box measuring 20" by 20" by 20" high which has been partitioned with corrigated cardboard into 36 uniform compartments.

5. The boxes will then be sealed in the standard manner ready for shipment.

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5.2 Packing, Level C.-Packing of the bands shall be in accordance with MIL-STD-1169.

5.3 Marking.-The contractor shall mark all shipments in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering Data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1395-A208

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. *If there are additional papers*, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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